

Dr. KAMAL PANDEY, PhD

Senior Staff Software architect | Agentic Applications & Enterprise LLM Platform Architecture

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PROFESSIONAL SUMMARY

Senior Staff software engineer and architect with 17+ years of designing and personally building large-scale distributed systems, including 5+ years of architecture and deploying complex agentic applications leveraging LLMs in production at enterprise scale. Serve as the technical authority for enterprise-wide agentic and LLM integration strategy at Rivian Automotive — setting architecture standards adopted across engineering organizations, partnering directly with senior leadership on a multi-year cognitive-automation roadmap, and acting as escalation point for architecture decisions spanning factory, supply-chain, and workforce systems. Deep, hands-on Python engineer with expert-level fluency in distributed, fault-tolerant, observable systems on AWS and Kubernetes. Track record of translating emerging machine-learning research into durable, company-wide technical strategy, and of mentoring and bar-raising engineers from mid-level through principal/staff. Fellow of IET, Senior Member of IEEE, published author of 20+ peer-reviewed papers and two books on agentic AI and digital transformation — sustained, industry-recognized technical impact beyond any single team.

CORE QUALIFICATIONS

- **Agentic Systems at Enterprise Scale:** 17+ years hands-on software engineering; 5+ years architecture and personally deploying complex agentic applications (Google A2A, CrewAI, Google ADK, LangChain, Amazon Bedrock) leveraging LLMs in production at enterprise scale.
- **Enterprise LLM Integration Strategy:** Own the multi-year technical vision for LLM integration across Rivian's software ecosystem — the final technical authority and escalation point for architecture decisions spanning factory, supply-chain, and workforce systems.
- **Cognitive Automation Roadmap:** Partner directly with senior leadership and product executives to define company-wide cognitive-automation direction, turning information extraction, summarization, and Q&A into scalable, repeatable enterprise capabilities.
- **Distributed Systems & Cloud:** Expert-level Python; working proficiency across TypeScript/Node.js and C#; deep fluency with AWS and Kubernetes for building fault-tolerant, observable distributed systems adopted org-wide.
- **Engineering Standards at Organizational Scale:** Established and enforced NIST RMF- and ISO 42001-anchored engineering standards for production-grade ML systems, driving consistent adoption across teams beyond my own.
- **ML Depth & Research Fluency:** Authoritative, research-grounded command of ML principles, algorithms, and evaluation metrics — 20+ peer-reviewed publications and two books translating state-of-the-art research into durable business impact.
- **Production Ownership at Scale:** Own design, deployment, and operation of mission-critical backend systems in high-stakes production environments at company-wide scale, including 100+ live agentic systems and \$100M+ in measured annual impact.
- **Technical Leadership & Talent Growth:** Proven ability to build consensus among executive stakeholders, mentor engineers from mid-level to principal/staff, and serve as technical bar-raiser — invited keynote speaker at Microsoft Ignite, Google Next, and Ai4.

PROFESSIONAL EXPERIENCE

Sr. Solutions Architect, Applied AI & Software Engineering — Rivian Automotive Inc. Feb 2021 – Present

- Serve as the preeminent technical authority for Rivian's agentic AI and GenAI platform, setting architecture and strategy spanning factory, supply-chain, engineering, and workforce organizations and shaping the multi-year technology roadmap.
- Architect and personally build mission-critical intelligent agent systems (Google A2A, Amazon Bedrock, CrewAI, Google ADK, LangChain) that automate workflows and elevate user experience — 100+ systems shipped, raising production defect detection from 40% to 97% and saving \$100M+ annually.

- Own the enterprise LLM integration strategy end-to-end: robust, scalable, maintainable data exchange across SAP, Workday, and Salesforce, acting as final escalation point for architecture decisions and eliminating manual handoffs for \$5M+ in annual savings.
- Partner directly with senior leadership to define and own the company-wide cognitive-automation roadmap, converting information extraction and document review into repeatable enterprise capability — automating 200+ compliance/legal processes and cutting cycles from 15 days to under 30 minutes.
- Design, build, and operate distributed, fault-tolerant, observable backend systems and APIs in Python and TypeScript/Node.js on Kubernetes/Terraform, cutting deployment time 70% while holding sub-30ms inference latency at enterprise scale.
- Set and enforce engineering standards for production-grade ML systems company-wide, anchored in NIST RMF and ISO 42001, including a repeatable framework for evaluating AI outputs and mitigating hallucination risk.
- Mentor and elevate engineers from mid-level through principal, lead code reviews across the AI/ML and platform organization and serve as a technical bar-raiser training co-workers on new architecture and tooling.
- Continuously research emerging agentic-AI and distributed-systems trends, translating findings into technical strategy with company-wide measurable impact; represent Rivian's technical point of view at Microsoft Ignite, Google Next, and Ai4.

Cloud Solutions Architect, Software Engineering — Samsung SDS Co., Ltd.

Mar 2020 – Mar 2021

- Personally designed and built distributed backend services and APIs for enterprise chatbots, including intent-matching and caching logic for low-latency response, achieving 70% query containment and \$300K in annual savings.
- Owned CI/CD and anomaly-detection pipelines (Jenkins, AWS) as production infrastructure, improving release throughput by 89% and codifying DevOps standards adopted team-wide.
- Led architecture for migrating 10,000+ legacy applications from Lotus Notes to Azure, modernizing distributed infrastructure and developer tooling for 65K+ global users.

Solutions Architect — The Goodyear Tire & Rubber Company

Aug 2015 – Oct 2019

- Personally designed and built a customer-facing AI chatbot and its backend APIs, translating business requirements directly into production software — 17% lift in sales conversion, \$2.3M in revenue growth.
- Delivered 23 custom distributed applications integrated with SAP and Power BI, writing the backend integration logic that eliminated third-party licensing costs and saved \$800K annually.

EARLY CAREER

- Technical Project Lead — Larsen & Toubro Infotech Ltd, USA (Mar 2012 – Jul 2015)
- Consultant — Singtel Optus Pty Limited, Sydney, Australia (Dec 2011 – Mar 2012)
- Lead Engineer — HCL Technologies Ltd, Singapore (May 2010 – Dec 2011)
- Senior Software Engineer — Shriram Value Services, Chennai, India (Feb 2008 – Apr 2010)

TECHNICAL SKILLS

- **Languages & Distributed Systems:** Python (expert), TypeScript/Node.js, C# — API design, distributed systems, data structures, and algorithms at production scale.
- **Agentic AI & LLM Platforms:** Prompt engineering, LangChain, LlamaIndex, CrewAI, Google ADK, Amazon Bedrock, Google A2A, Vector Databases (Pinecone, ChromaDB), Knowledge Graphs, RAG frameworks, PyTorch, TensorFlow.
- **Cloud, Containers & Observability:** AWS (SageMaker, Lambda), Kubernetes, Docker, Terraform, GCP (Vertex AI, BigQuery), Azure (AKS, Synapse) — fault-tolerant, observable systems at organizational scale.
- **Governance & ML Quality:** NIST RMF, ISO 42001, Responsible AI Lifecycle (SAIL), IAM, VPC security, AI-output validation and hallucination mitigation frameworks.

EDUCATION & CERTIFICATIONS

- Master's in Computer Science — I.K. Gujral Punjab Technical University
- Ph.D. in Information Systems — Dakota State University (2025–2029, in progress)
- DBA, Computer Science — Westcliff University, USA (2023)
- 30+ certifications including TOGAF, SAFe 6 Architect, PMP, Azure & Google AI Solutions Architect, ITIL Certified Professional.

AWARDS, PUBLICATIONS & AFFILIATIONS

- Global AI Leader 2025 and 2026 — Constellation Research Inc.
- Top 50 AI Research Reviewer Award — Stanford University, 2025.
- Keynote speaker: Microsoft Ignite (2016, 2018), Google Next (2025), Ai4 (2025).
- 20+ peer-reviewed publications; reviewed 50+ research papers; author of two books on AI, ML, and agentic/digital transformation.
- Fellow of IET, Senior Member of IEEE, Member of AAAI.
- Authors three AI/ML books published by Pocket